

Improving Data Efficiency in Agricultural Field Segmentation Using Dataset Scaling and Active Learning Strategies

Phuong Uyen (Luna) Nguyen | Computer Systems Engineering, Arizona State University
Faculty Mentor: Dr. Hannah Kerner | School of Computing and Augmented Intelligence (SCAI)

The research question this project seeks to address is: *How does training dataset size affect agricultural field segmentation performance, and can simple active learning strategies reduce the amount of labeled data needed while maintaining high accuracy?* This project will measure how model accuracy changes as we use more or less labeled training data, and test whether a smarter way of picking which samples to label can give the same results with less work. We hypothesize two main outcomes. First, model performance will improve rapidly with increasing dataset size initially, then plateau beyond a certain threshold, indicating diminishing returns in labeling effort. Second, we will evaluate whether uncertainty-based active learning sampling can achieve accuracy comparable to random sampling while requiring fewer labeled samples, thereby reducing the overall data annotation burden. The primary focus of this project will be the dataset scaling experiments, with active learning explored as an extension if time permits.

This research advances knowledge in the area of data-efficient machine learning for remote sensing. This project aligns with the Fulton Schools' **Data research theme**, which focuses on new computational methods, machine learning, and data-driven decision-making. Labeling satellite imagery for agricultural field segmentation is expensive and slow, which limits how widely these systems can be deployed. The Fields of the World (FTW) dataset, developed by Dr. Kerner and collaborators at ASU and Microsoft, is a large benchmark with 70,462 labeled samples from 24 countries using Sentinel-2 satellite images [1]. While FTW showed that models trained on diverse regions generalize well, it is still unclear how much labeled data is actually needed for reliable performance. U-Net [2], a widely used encoder-decoder segmentation network, is the baseline architecture for this project because it performs well on pixel-level tasks like field boundary detection. Active learning [3] is a technique where a model selects the most uncertain unlabeled samples to be labeled next, repeating this process to learn effectively with less data. Recent work has shown that uncertainty-based active learning can reduce the number of annotations needed in segmentation tasks [4], though results depend on the dataset [5]. This project will measure these trade-offs directly using the FTW data.

In order to address the research question and hypothesis described above, the following research plan is proposed. The study will use the publicly available FTW dataset [1] with a U-Net model [2] implemented in Python using PyTorch. All hyperparameters (learning rate, batch size, epochs) will be held constant across all experiments. The research has two main parts: (1) dataset scaling, training the same model on increasing amounts of labeled data to measure how performance changes, and (2) an active learning comparison, testing a smarter sampling strategy against random sampling at the same data budgets.

Materials

- Dataset: Fields of the World (FTW) [1] (publicly available, no cost)
- Software: Python, PyTorch, scikit-learn for model implementation and evaluation
- Hardware: Personal computer with GPU support

Participants

This study is entirely computational and does not involve human participants.

Facilities

ASU Research Computing will be used for GPU-accelerated model training if personal hardware is insufficient.

Procedures

1. Set up the Python/PyTorch development environment and reproduce the FTW U-Net [2] baseline to confirm correct implementation.
2. **Dataset Scaling:** Randomly subsample the FTW training set to 10%, 20%, 40%, 60%, 80%, and 100%.
3. Train a separate U-Net model for each subset under identical hyperparameter settings (learning rate, batch size, epochs).
4. Evaluate each model on a fixed held-out test set using Intersection over Union (IoU), precision, and recall.
5. Plot learning curves (IoU vs. dataset size) to identify where performance stops improving significantly.
6. **Active Learning:** Train an initial model on the 10% seed set.
7. Score all remaining unlabeled samples by prediction entropy (model uncertainty).
8. Select the most uncertain samples, add them to the training set, and retrain the model.
9. Repeat Steps 7-8 until the same total data budget as the scaling experiments is reached.
10. Compare active learning vs. random sampling efficiency curves (IoU vs. labeled samples).

Expected Outcomes

Scaling experiments: Learning curves showing where IoU plateaus, identifying the minimum labeled data needed for reliable field segmentation.

Active learning comparison: Efficiency curves for both strategies showing whether uncertainty sampling reaches the same accuracy with fewer labels.

Analysis: IoU scores compared across all conditions; labeling savings from active learning calculated at equivalent accuracy levels.

References

- [1] H. Kerner et al., "Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 27, pp. 28151-28159, 2025. doi: 10.1609/aaai.v39i27.35034
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2015*, Lecture Notes in Computer Science, vol. 9351, Springer, Cham, 2015, pp. 234-241. doi: 10.1007/978-3-319-24574-4_28
- [3] B. Settles, "Active Learning Literature Survey," Computer Sciences Technical Report 1648, University of Wisconsin-Madison, 2009. [Online]. Available: <https://burrsettles.com/pub/settles.activelearning.pdf>
- [4] Z. Hu et al., "A Survey on Deep Active Learning: Recent Advances and New Frontiers," *arXiv preprint*, arXiv:2405.00334, 2024. [Online]. Available: <https://arxiv.org/abs/2405.00334>
- [5] A. Bengar et al., "Realistic Evaluation of Deep Active Learning for Image Classification and Semantic Segmentation," *International Journal of Computer Vision*, 2025. doi: 10.1007/s11263-025-02372-z

Timeline

Month	Phase	Dates	Tasks
Aug 2026	Research Setup	Aug 24 - Aug 28 (Week 1)	Conduct literature review on data efficiency and agricultural field segmentation; read FTW paper in depth
		Aug 31 - Sep 4 (Week 2)	Set up Python/PyTorch environment; reproduce FTW U-Net baseline and verify performance
Sep 2026	Dataset Scaling Experiments	Sep 7 - Sep 11 (Week 3)	Prepare FTW dataset subsets: 10%, 20%, 40%, 60%, 80%, 100%
		Sep 14 - Sep 25 (Weeks 4-5)	Train U-Net models on each subset under consistent conditions; log IoU, precision, recall
		Sep 28 - Oct 2 (Week 6)	Plot learning curves; analyze performance vs. dataset size results
Oct 2026	Active Learning Impl.	Oct 5 - Oct 9 (Week 7)	Implement uncertainty-based sampling pipeline (entropy scoring of unlabeled samples)
		Oct 12 - Oct 23 (Weeks 8-9)	Run iterative active learning experiments; compare against random sampling at equal data budgets

		Oct 26 - Oct 30 (Week 10)	Generate active learning efficiency curves; begin cross-experiment analysis
Nov 2026	Evaluation and Analysis	Nov 2 - Nov 13 (Weeks 11-12)	Compare all experiments; identify dataset size threshold and labeling savings from active learning
		Nov 16 - Nov 20 (Week 13)	Finalize all plots, tables, and analysis; begin writing final report
Dec 2026	Finalization and Presentation	Nov 30 - Dec 11 (Weeks 14-15)	Complete final report and FURI Fulton Forge poster; submit all deliverables
		Dec 2026 (Week 16)	Prepare and rehearse FURI Symposium presentation

Budget

No specialized hardware purchases are required. All experiments will be conducted using personal computing resources or university-provided compute resources (ASU Research Computing). The FTW dataset is publicly available at no cost. No supply budget is requested at this time.

Personal Statement

Phuong Uyen (Luna) Nguyen | Computer Systems Engineering, Arizona State University

Growing up watching agricultural landscapes in Vietnam and later witnessing drought-driven crop failures reported in the news, I came to understand early that food security is not an abstract policy problem; it is something that touches families directly. As a Computer Systems Engineering student at Arizona State University, I am drawn to research that uses computational tools to address that kind of tangible, global challenge. The opportunity to work with Dr. Hannah Kerner on data-efficient methods for agricultural field segmentation represents exactly the kind of work I want to pursue.

My interest in machine learning applied to real-world systems has deepened steadily throughout my time at ASU. In my current course, CSE 475 (Foundations of Machine Learning), I am building both the theoretical grounding and practical skills to design, train, and evaluate ML models. This complements my earlier hands-on experience from EGR 102, where I led the programming for an Arduino-based rain gauge system that won the "Most Innovative/Creative" award, a project that first showed me how sensor data and intelligent algorithms could produce meaningful environmental insights at low cost. My work as a Documentation Lead in the EPICS PowerPaws Service Dog Grocery Store Shopping project further developed my ability to manage deliverables in research and development settings.

The FURI project I am proposing builds directly on Dr. Kerner's Fields of the World (FTW) research, which she recommended as the starting point in her response to my mentorship inquiry. I designed this proposal around the specific directions she outlined: investigating how model performance changes as dataset size varies from 10% to 100% of available samples, and comparing uncertainty-based active learning sampling against random sampling to determine whether intelligent sample selection can achieve comparable accuracy with less labeled data. These are exactly the kinds of practical data efficiency questions that matter for real-world deployment of agricultural ML systems in regions where labeling resources are scarce.

My goal in pursuing FURI is to develop genuine research skills under faculty mentorship, gain experience in the full ML research cycle from hypothesis through evaluation, and contribute meaningfully to a project with real global impact. I am committed to completing all planned experiments, producing rigorous analysis, and presenting my findings at the Fulton Forge Student Research Expo. In the longer term, this experience will inform my path toward graduate school and a career at the intersection of machine learning and environmental systems.

I am excited about the opportunity FURI provides and am grateful for Dr. Kerner's willingness to mentor this project. I am ready to work hard, learn quickly, and contribute to research that matters.

PHUONG UYEN (LUNA) NGUYEN

+1 (253) 329-4776 • pnguye79@asu.edu • [LinkedIn](#)

EDUCATION

B.S.E., Computer Systems Engineering Expected May 2027
Arizona State University, Tempe, AZ 4.0 GPA

- Awards/Scholarships: New American University - University Award

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB
AI / Machine Learning: TensorFlow, CNN training, OpenCV, image preprocessing
Systems & Tools: Linux/Ubuntu, Git/GitHub, Firebase
Hardware & Embedded: Raspberry Pi, Arduino, FPGA
Concepts: Data processing pipelines, automation scripting, computer vision, embedded systems, debugging

RESEARCH & PROJECT EXPERIENCE

AI-Powered Adaptive Robotic Arm - Research Aide January 2025 - May 2025

Fulton Undergraduate Research Initiative (FURI), Arizona State University

- Developed an intelligent robotic manipulation system using Python, OpenCV, and Raspberry Pi
- Built computer vision pipeline performing real-time object detection with ~87% accuracy across varying lighting conditions
- Trained CNN models in TensorFlow/Keras using 1000+ labeled samples to improve object recognition
- Integrated machine learning inference with hardware control to enable adaptive pick-and-place tasks

Real-Time Services Capacity Sensor & Alert System - Team Lead

EPICS Engineering Program, Arizona State University August 2025 - December 2025

Community Partner: Florida Community Innovation (FCI)

- Led a 7-member interdisciplinary engineering team building a real-time shelter capacity tracking prototype
- Designed system architecture integrating barcode scanning hardware with Firebase cloud database, creating a scalable low-cost solution adaptable to multiple sites
- Developed Python data processing scripts in a Linux (Ubuntu) environment to support backend data handling and system validation
- Coordinated user research, design documentation, and mentor feedback sessions to align system design with real stakeholder workflows.

WORK & LEADERSHIP EXPERIENCE

Department Administrative Support Assistant, September 2025 – Present
EdPlus Enrollment Center, Arizona State University, Scottsdale, Arizona

- Organized and maintained departmental data across Google Suite and SharePoint systems, improving record accuracy and reducing report preparation time by ~20%.

Director of Outreach, *Connecting International Communities (CIC)*,

Arizona State University, Tempe, Arizona January 2024 - May 2025

- Recruited and led a 25-member volunteer team to coordinate cultural and orientation events, increasing student participation by 300+ members.

Arizona State University
Unofficial Transcript

Name: Phuong Uyen Nguyen
Student ID: 1229611684

Print Date: 03/18/2026

Transfer Credits					
Transfer Credit from Rio Salado College					
			<u>Attempted</u>	<u>Earned</u>	<u>Points</u>
Course Trans	0.000	Transfer Totals:	0.000	7.000	0.000
GPA:					
Transfer Credit from Chandler-Gilbert Community College					
			<u>Attempted</u>	<u>Earned</u>	<u>Points</u>
Course Trans	0.000	Transfer Totals:	0.000	3.000	0.000
GPA:					
Beginning of Undergraduate Record					

2023 Fall					
<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>	<u>Points</u>
ASU 101-TPS	The ASU Experience	1.000	1.000	A+	4.333
EGR 101	Found of Eng Design Proj I	3.000	3.000	A+	12.999
ENG 107	First-Year Composition	3.000	3.000	A+	12.999
FIN 123	ASU Financial Literacy	1.000	1.000	A	4.000
HSE 101	Human Systems Engineering	3.000	3.000	A+	12.999
MAT 265	Calculus for Engineers I	3.000	3.000	A+	12.999
PHI 105	Intro to Ethics	3.000	3.000	A-	11.001
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	4.00	Term Totals	17.000	17.000	71.330
Cum GPA:	4.00	Cum Totals	17.000	17.000	71.330
Academic Standing: Good Standing					
Term Honor: Dean's List					

2024 Spring					
<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>	<u>Points</u>
CHM 113	General Chemistry I	4.000	4.000	A+	17.332
EGR 102	Found of Eng Design Proj II	3.000	3.000	A+	12.999
EGR 104	Crit Inquiry in Engineering	3.000	3.000	A	12.000
ENG 108	First-Year Composition	3.000	3.000	A+	12.999
FSE 104	EPICS Gold I	1.000	1.000	A	4.000
MAT 266	Calculus for Engineers II	3.000	3.000	A	12.000
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	4.00	Term Totals	17.000	17.000	71.330
Cum GPA:	4.00	Cum Totals	34.000	34.000	142.660
Academic Standing: Good Standing					
Term Honor: Dean's List					

2024 Fall

<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>	<u>Points</u>
EGR 201	Use-Inspired Design Project I	3.000	3.000	A+	12.999
EGR 216	EGR Electrical Fundamentals	3.000	3.000	A	12.000
EGR 218	Materials and Manufact Process	3.000	3.000	A	12.000
MAT 267	Calculus for Engineers III	3.000	3.000	A	12.000
PHY 121	Univ Physics I: Mechanics	3.000	3.000	A	12.000
PHY 122	University Physics Lab I	1.000	1.000	A	4.000
TEM 200	My Technology Venture	3.000	3.000	A	12.000
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	4.00	Term Totals	19.000	19.000	76.999
Cum GPA:	4.00	Cum Totals	53.000	56.000	219.659

Academic Standing: Good Standing
Term Honor: Dean's List

2025 Spring					
<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>	<u>Points</u>
CSE 205	Object-Oriented Program & Data	3.000	3.000	A	12.000
EEE 120	Digital Design Fundamentals	3.000	3.000	A+	12.999
MAT 243	Discrete Math Structures	3.000	3.000	A	12.000
MAT 275	Modern Differential Equations	3.000	3.000	A	12.000
PHY 131	Univ Physics II: Elctrc/Magnet	3.000	3.000	A+	12.999
PHY 132	University Physics Lab II	1.000	1.000	A+	4.333
PUP 200	Cities in Cinema	3.000	3.000	A+	12.999
WST 100	Women, Gender, and Society	3.000	3.000	A+	12.999
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	4.00	Term Totals	22.000	22.000	92.329
Cum GPA:	4.00	Cum Totals	75.000	78.000	311.988
Academic Standing: Good Standing					
Term Honor: Dean's List					

2025 Fall					
<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>	<u>Points</u>
CSE 301	Computing Ethics	1.000	1.000	A+	4.333
CSE 310	Data Structures and Algorithms	3.000	3.000	A	12.000
CSE 320	Design/Syn Digital Hardware	3.000	3.000	A	12.000
EEE 202	Circuits I	4.000	4.000	A+	17.332
FSE 404	EPICS Gold: EPICS in Action	1.000	1.000	A	4.000
IEE 380	Prob & Stats Engr Prob Solving	3.000	3.000	A	12.000
MAT 343	Applied Linear Algebra	3.000	3.000	A+	12.999
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	4.00	Term Totals	18.000	18.000	74.664

Arizona State University

Unofficial Transcript

Name: **Phuong Uyen Nguyen**
Student ID: **1229611684**

Cum GPA: 4.00 Cum Totals 93.000 103.000 386.652

Academic Standing: Good Standing
Term Honor: Dean's List

2026 Spring

Course	Description	Attempted	Earned	Grade	Points
CSE 325	Embedded Microprocessor System	3.000	0.000	NR	0.000
CSE 330	Operating Systems	3.000	0.000	NR	0.000
CSE 360	Intro to Software Engineering	3.000	0.000	NR	0.000
CSE 420	Computer Architecture I	3.000	0.000	NR	0.000
CSE 463	Intro Human Computer Interactn	3.000	0.000	NR	0.000
CSE 475	Found of Machine Learning	3.000	0.000	NR	0.000
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	0.00	Term Totals	0.000	0.000	0.000
Cum GPA:	4.00	Cum Totals	93.000	103.000	386.652

2026 Fall

Course	Description	Attempted	Earned	Grade	Points
CSE 412	Database Management	3.000	0.000	NR	0.000
CSE 423	Systems Capstone Project I	3.000	0.000	NR	0.000
CSE 472	Social Media Mining	3.000	0.000	NR	0.000
EEE 334	Circuits II	4.000	0.000	NR	0.000
FSE 301	Entrepreneur & Value Creation	3.000	0.000	NR	0.000
SWU 349	Stress Management Tools II	3.000	0.000	NR	0.000
		<u>Attempted</u>	<u>Earned</u>		<u>Points</u>
Term GPA:	0.00	Term Totals	0.000	0.000	0.000
Cum GPA:	4.00	Cum Totals	93.000	103.000	386.652

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